Ratio, Root, and Alternating Series Tests

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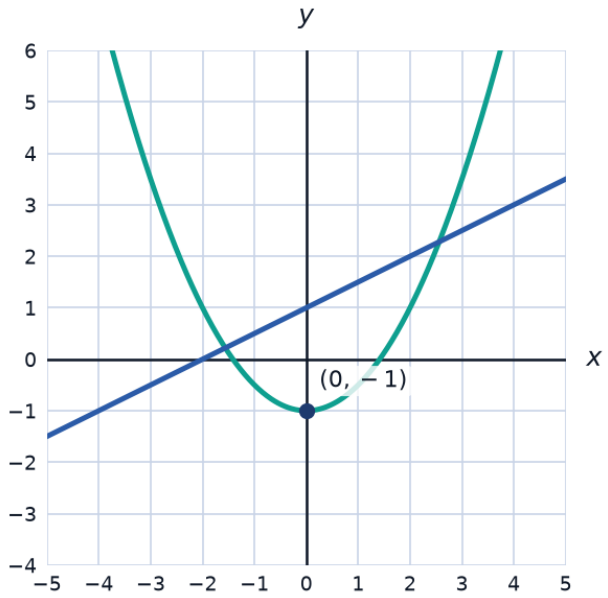
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Ratio Test**

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

2 **[Foundational]** • **Target: The Ratio Test**

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{3^n}$ converges.



3 **[Foundational]** • **Target: The Root Test**

Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.

4 **[Foundational]** • **Target: Alternating series and the Alternating Series Test**

State the Alternating Series Test, and use it to show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.

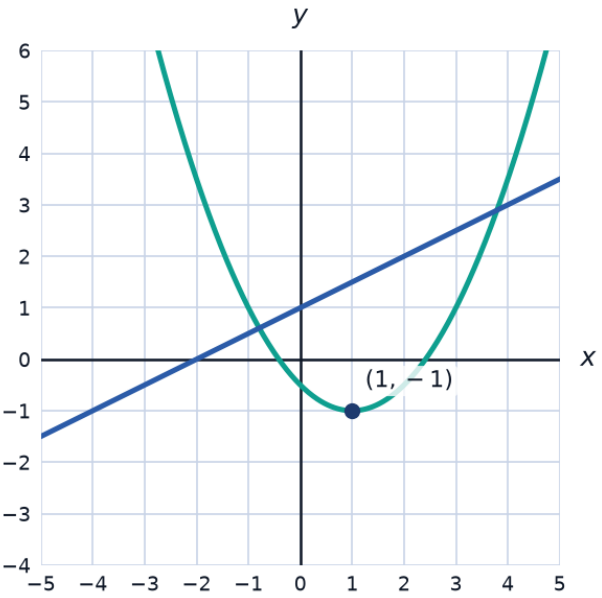
5 **[Foundational]** • **Target: Alternating series and the Alternating Series Test**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ converges using the Alternating Series Test.

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6 [Foundational] • Target: Alternating series error bound

Approximate $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ using the first 4 terms, and bound the error using the Alternating Series Error Bound.



7 [Foundational] • Target: Absolute vs. conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges absolutely, conditionally, or diverges.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Absolute vs. conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges absolutely, conditionally, or diverges.

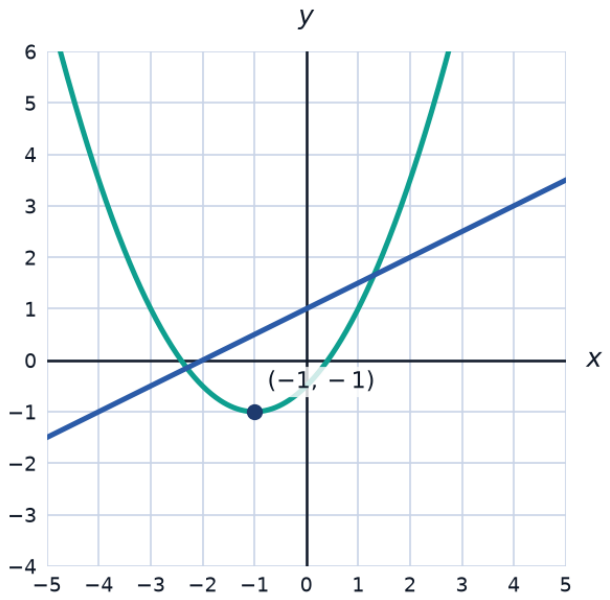
9 [Intermediate] • Target: More Ratio Test practice

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges.

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10 [Intermediate] • Target: More Ratio Test practice

Explain why the Ratio Test is INCONCLUSIVE for $\sum_{n=1}^{\infty} \frac{1}{n}$, and identify which test should be used instead.



11 [Intermediate] • Target: Partial sums of an alternating series (visualizing oscillating convergence)

Compute the first 5 partial sums of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ (the alternating harmonic series, which converges to $\ln 2 \approx 0.693$), and describe how they oscillate.

12 [Intermediate] • Target: Determining the number of terms needed for a given accuracy

For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$, find the minimum number of terms needed so the partial sum is within 0.001 of the true sum, using the Alternating Series Error Bound.

13 [Intermediate] • Target: Mixed practice: classify each series

Classify each series as convergent (state absolutely or conditionally) or divergent:

a $\sum \frac{(-1)^n}{\sqrt{n}}$

b $\sum \frac{(-1)^n n^2}{n^3+1}$

c $\sum \frac{(-1)^n}{n!}$

14 [Intermediate] • Target: The Root Test with more complex expressions

Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$ converges.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

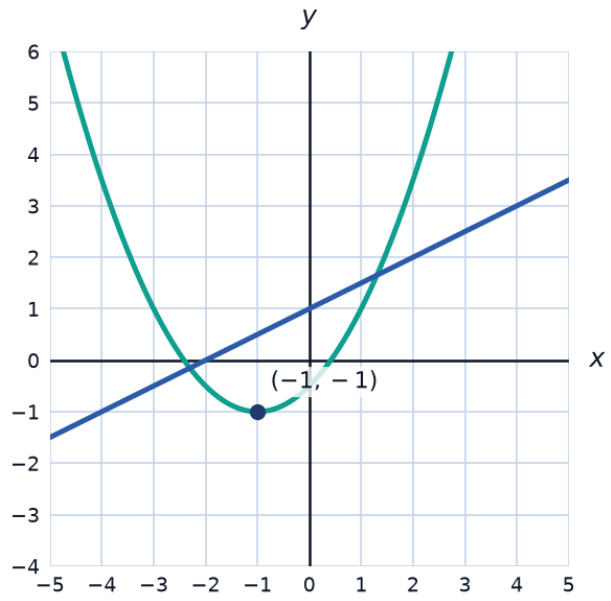
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15 [Challenging] • Target: The Root Test with more complex expressions • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Use the Root Test to determine whether $\sum_{n=1}^{\infty} \frac{2^n}{n^n}$ converges.

16 [Challenging] • Target: Why the Ratio Test fails for p-series • Cross-Lesson: 3.01 Parametric Equations

Show that the Ratio Test is inconclusive for EVERY p-series $\sum \frac{1}{n^p}$, regardless of p.



17 [Challenging] • Target: Combining multiple tests in one problem • Cross-Lesson: 3.02 Vector-Valued Functions

Determine whether $\sum_{n=1}^{\infty} \frac{(-2)^n}{n^2}$ converges absolutely, conditionally, or diverges.

18 [Challenging] • Target: Alternating series and the Alternating Series Test • Cross-Lesson: 4.01 Polar Curves and Derivatives

For $x = 3t,\ y = t^2 - 1$, find $\frac{dy}{dx}$.

19 [Challenging] • Target: Alternating series error bound • Cross-Lesson: 4.02 Area in Polar Coordinates

Convert the rectangular point $(1, 1)$ to polar coordinates.

20 [Challenging] • Target: Absolute vs. conditional convergence • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Does $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ converge?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Ratio Test

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

a. 3

b. $\frac{1}{3}$

c. $\frac{4}{3}$

d. $\frac{1}{4}$

22 [Foundational] • Target: The Root Test

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{14^n}$.

a. 14

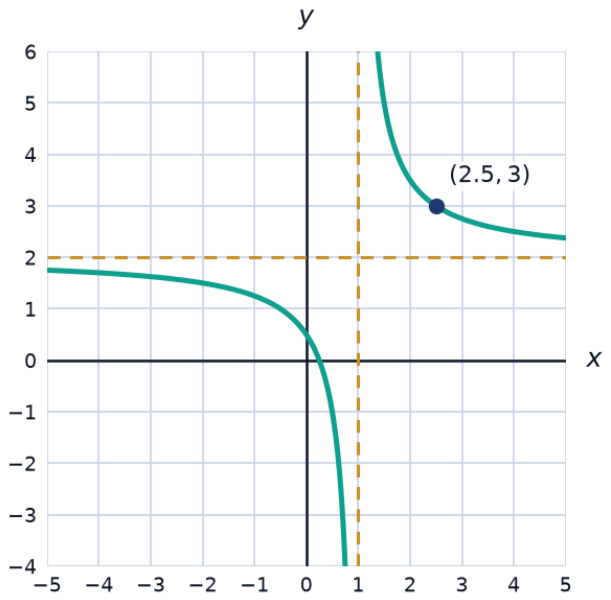
b. $\frac{1}{14}$

c. $\frac{1}{15}$

d. $\frac{15}{14}$

23 [Foundational] • Target: Alternating series and the Alternating Series Test

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{10^n}$.



a. 10

b. $\frac{1}{11}$

c. $\frac{1}{10}$

d. $\frac{11}{10}$

24 [Foundational] • Target: Alternating series error bound

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{13^n}$.

a. $\frac{1}{14}$

b. 13

c. $\frac{14}{13}$

d. $\frac{1}{13}$

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25 [Foundational] • Target: Absolute vs. conditional convergence

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{11^n}$.

a. $\frac{1}{11}$

b. $\frac{1}{12}$

c. 11

d. $\frac{12}{11}$

26 [Foundational] • Target: More Ratio Test practice

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{12^n}$.

a. $\frac{13}{12}$

b. $\frac{1}{13}$

c. $\frac{1}{12}$

d. 12

27 [Foundational] • Target: Partial sums of an alternating series (visualizing oscillating convergence)

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{16^n}$.

a. $\frac{1}{16}$

b. $\frac{1}{17}$

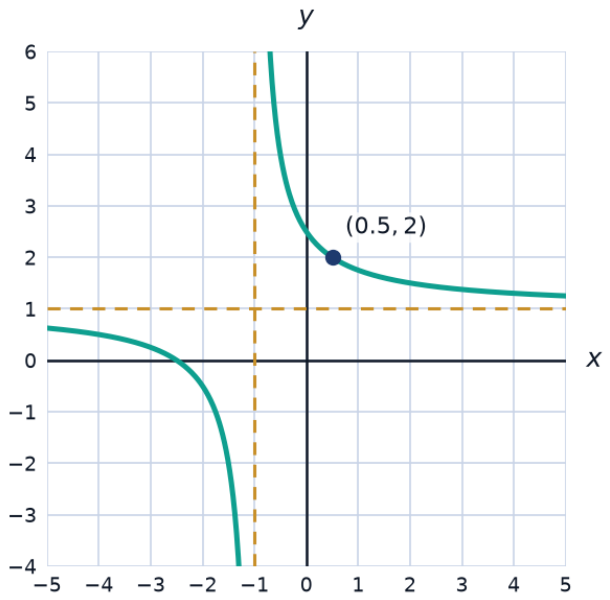
c. 16

d. $\frac{17}{16}$

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28 [Foundational] • Target: Determining the number of terms needed for a given accuracy

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{4n+4}{11n-7}\right)^n\right|}$.



a. $\frac{4}{11}$

b. $\frac{11}{4}$

c. $\frac{19}{22}$

d. $\frac{15}{11}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mixed practice: classify each series

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{7n-7}{12n+4}\right)^n\right|}$.

a. $\frac{12}{7}$

b. $\frac{7}{12}$

c. $\frac{19}{12}$

d. $\frac{13}{12}$

30 [Intermediate] • Target: The Root Test with more complex expressions

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{4n+6}{9n-2}\right)^n\right|}$.

a. $\frac{9}{4}$

b. $\frac{4}{9}$

c. $\frac{17}{18}$

d. $\frac{13}{9}$

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31 **[Intermediate]** • **Target: Why the Ratio Test fails for p-series**

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{9n+4}{12n+3}\right)^n\right|}$.

a. $\frac{3}{4}$ **b.** $\frac{7}{4}$

c. $\frac{4}{3}$ **d.** $\frac{5}{4}$

32 **[Intermediate]** • **Target: Combining multiple tests in one problem**

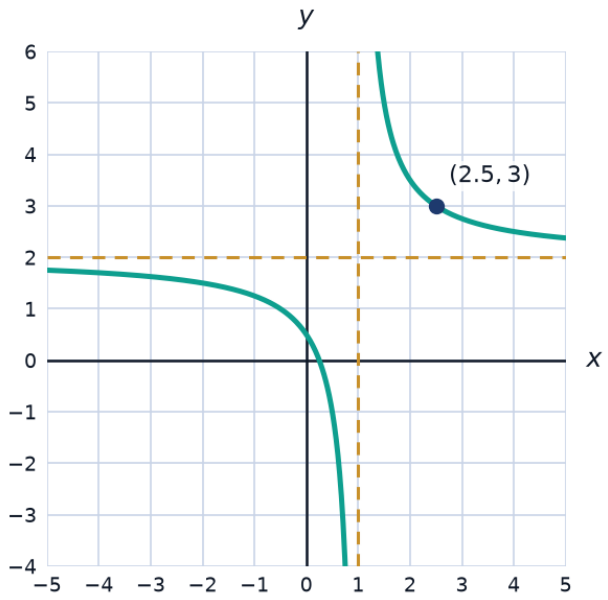
Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{6n-6}{12n-3}\right)^n\right|}$.

a. $\frac{3}{2}$ **b.** $\frac{1}{2}$

c. 2 **d.** 1

33 **[Intermediate]** • **Target: The Alternating Series Test**

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{8n-6}{13n+7}\right)^n\right|}$.



a. $\frac{21}{13}$ **b.** $\frac{8}{13}$

c. $\frac{29}{26}$ **d.** $\frac{13}{8}$

34 **[Intermediate]** • **Target: Absolute vs. Conditional Convergence**

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{8n-5}{12n-4}\right)^n\right|}$.

a. $\frac{3}{2}$ **b.** $\frac{2}{3}$

c. $\frac{5}{3}$ **d.** $\frac{7}{6}$

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35 **[Intermediate]** • **Target: Alternating Series Error Bound**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+6}$ converges absolutely, conditionally, or diverges.

a. Cannot be determined without more information

b. Diverges

c. Conditionally convergent

d. Absolutely convergent

36 **[Intermediate]** • **Target: The Ratio Test**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 4}{n^1}$ converges absolutely, conditionally, or diverges.

a. Absolutely convergent

b. Cannot be determined without more information

c. Diverges

d. Conditionally convergent

37 **[Intermediate]** • **Target: The Root Test**

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{8^n}{n!}$ converges.

a. Diverges ($L = \infty$)

b. Converges by the Alternating Series Test

c. Inconclusive ($L = 1$)

d. Converges ($L = 0$)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Alternating series and the Alternating Series Test • Cross-Lesson: 4.01 Polar Curves and Derivatives

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{7^n}{n!}$ converges.

a. Inconclusive ($L = 1$)

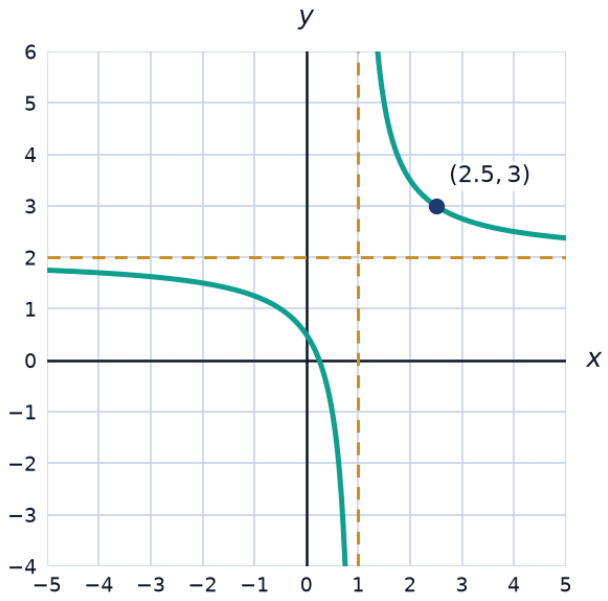
b. Converges ($L = 0$)

c. Diverges ($L = \infty$)

d. Converges by the Alternating Series Test

39 [Challenging] • Target: Alternating series error bound • Cross-Lesson: 4.02 Area in Polar Coordinates

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left(\frac{3n}{n+5}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{3n}{n+5}\right)^n$.



a. 4

b. 2

c. 5

d. 3

40 [Challenging] • Target: Absolute vs. conditional convergence • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left(\frac{9n}{n+1}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{9n}{n+1}\right)^n$.

a. 1

b. 8

c. 10

d. 9

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41 [Challenging] • Target: The Ratio Test • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Evaluate the limit $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for the series $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

a. 0 (converges)

b. 3 (diverges)

c. 1 (inconclusive)

d. ∞ (diverges)

42 [Challenging] • Target: The Alternating Series Test • Cross-Lesson: 6.02 Taylor and Maclaurin Series

What two conditions must an alternating series $\sum (-1)^{n-1} b_n$ with $b_n > 0$ satisfy to converge by the AST?

a. $b_{n+1} \le b_n$ for all n, and $\lim_{n\to\infty} b_n = 0$

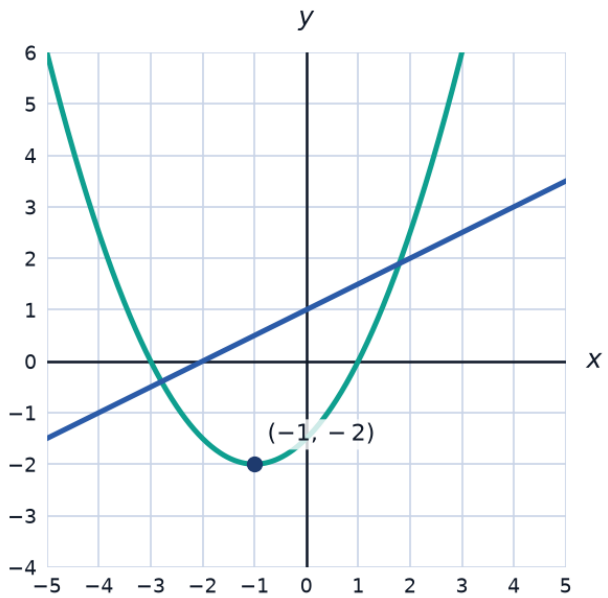
b. b_n is increasing, and $\lim_{n\to\infty} b_n = 1$

c. $\lim_{n\to\infty} b_n = 0$ only

d. The Ratio Test limit must be < 1

43 [Challenging] • Target: The Alternating Series Test • Cross-Lesson: 1.01 Integration by Parts

Determine the convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$.



a. Converges conditionally

b. Converges absolutely

c. Diverges

d. Inconclusive

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44 [Challenging] • Target: Absolute vs. Conditional Convergence • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Which of the following series converges absolutely?

a. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

b. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

c. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} (-1)^n$

45 [Challenging] • Target: Alternating Series Error Bound • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

For a convergent alternating series $\sum(-1)^{n-1}b_n$, what is the upper bound for the error $|R_n| = |S - S_n|$?

a. $|R_n| \le b_{n+1}$

b. $|R_n| \le b_n$

c. $|R_n| \le \frac{b_{n+1}}{2}$

d. $|R_n| \le \frac{b_n}{n}$

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